Province of the
EASTERN CAPE
EDUCATION

NATIONAL SENIOR CERTIFICATE

GRADE 10

NOVEMBER 2017

AGRICULTURAL SCIENCES P1 MARKING GUIDELINE

MARKS: 150

This marking guideline consists of 10 pages.

SECTION A

QUESTION 1

1.1	1.1.1	B ✓✓		
	1.1.2	B ✓✓		
	1.1.3	D ✓✓		
	1.1.4	C ✓✓		
	1.1.5	D ✓✓		
	1.1.6	C ✓✓		
	1.1.7	D ✓✓		
	1.1.8	B ✓✓		
	1.1.9	C ✓✓		
	1.1.10	B ✓✓	(10 x 2)	(20)
1.2	1.2.1	B only ✓✓		
	1.2.2	None ✓✓		
	1.2.3	Both A and B ✓✓		
	1.2.4	A only ✓✓		
	1.2.5	B only ✓✓	(5 x 2)	(10)
1.3	1.3.1	Palatable ✓✓		
	1.3.2	Competition ✓✓		
	1.3.3	Industrialisation ✓✓		
	1.3.4	Layers ✓✓		
	1.3.5	Poaching ✓✓	(5 x 2)	(10)

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1.4	1.4.1	Primary agriculture ✓		
	1.4.2	Arid/dry/unfavourable ✓		
	1.4.3	Capital ✓		
	1.4.4	Evaporation ✓		
	1.4.5	Foreign currency/foreign income ✓	(5 x 1)	(5)

TOTAL SECTION A: 45

SECTION B

QUESTION 2: AGRO-ECOLOGY

2.1 **Identification of the bacteria:**

2.1.1 (a) Denitrifying bacteria ✓ (1)
(b) Nitrifying bacteria ✓ (1)
(c) Nitrogen-fixing bacteria ✓ (1)

2.1.2 **Ways that a farmer can increase the nitrogen levels in the soil:**

- Adding nitrogen fertilisers to the soil ✓
- Adding organic matter such as compost or manure to the soil ✓
- Planting legumes that have nitrogen-fixing bacteria in their roots ✓ (Any 2 x 1) (2)

2.1.3 **Plants that have nodules in their roots:**

- Legumes ✓✓ (2)

2.2 **Organic farming**

2.2.1 **Crops produced by organic farmers from the scenario**

- Cereals ✓
- Vegetables ✓
- Roots and tubers ✓
- Herbs and spices ✓
- Fruits ✓
- Nuts ✓
- Rooibos tea ✓ (Any 2 x 1) (2)

2.2.2 **Farming methods that organic farmers use**

- Crop rotation ✓
- Green manure ✓
- Compost ✓
- Biological pest control ✓
- Mechanical weed control ✓ (Any 3 x 1) (3)

2.2.3 **Reasons why organic farming has become so popular the past couple of decades**

- Farming practices that do not disturb the environment ✓
- Less of a negative effect on biodiversity ✓
- Less pollution in soil and water ✓
- Reduce pesticides, growth hormones, antibiotics and preservatives ✓
- Healthier food produced that lessens risk of diseases ✓
- Fresh food with higher nutrient value and is tastier ✓

(Any 2 x 1) (2)

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2.3 **Symbiotic relationships found in ecosystems**

2.3.1 **Symbiotic relationships illustrated in diagrams A–C**
A – Competition ✓
B – Commensalism ✓
C – Mutualism ✓ (3)

2.3.2 **Other symbiotic relationships that are not illustrated in the diagram in QUESTION 2.3**
- Parasitism ✓
- Predator and prey ✓ (2)

2.4 **Examples of edaphic factors**
- Soil texture ✓
- Soil depth ✓
- Soil type ✓
- Soil water ✓
- Soil fertility ✓ (Any 3 x 1) (3)

2.5 **Table showing biomes**

2.5.1 **Correct answers for letter (A–F)**
A – Forest ✓
B – Large trees ✓
C – Sheep, cattle, goats ✓
D – Central plateau of South Africa, KwaZulu-Natal, Eastern Cape ✓
E – Savanna ✓
F – Low trees and grass ✓ (6)

2.5.2 Fynbos ✓ (1)

2.6 **Climate change**

2.6.1 **Four main greenhouse gases:**
- Water vapour ✓
- Carbon dioxide ✓
- Methane ✓
- Nitrous oxide ✓ (4)

2.6.2 **Human activities that cause an increase in greenhouse gases:**
- Industrial revolution ✓
- Green revolution ✓ (2)

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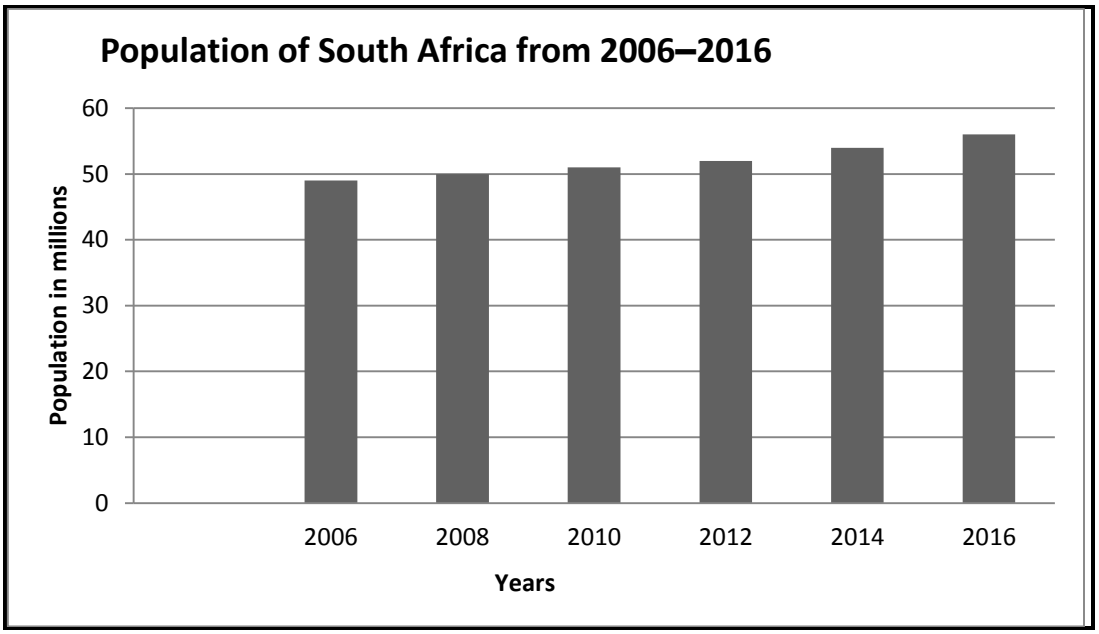
QUESTION 3: AGRI-INDUSTRY

3.1 Rounded-off figures of the population of South Africa from 2006–2016

3.1.1 Bar graph showing population of South Africa from 2006–2016



Criteria/rubric/marking guidelines

- Correct heading ✓
- Y-axis – correct label (Population in millions) ✓
- X-axis – correct label (Years) ✓
- Accuracy ✓
- Bar graph ✓
- Units (millions) ✓ (6)

3.1.2 Impacts of population growth on the agricultural sector

- Demand for food increases ✓
- Deforestation rates increase as more people need fuel wood ✓
- Farms become more mechanised ✓
- More people move to urban areas in search of employment opportunities ✓
- Use of agricultural land for housing ✓
- Agricultural land for food products decreases ✓
- Natural resources are degraded and depleted ✓
- Grazing land degenerates, which impacts negatively on livestock ✓ (Any 3 x 1) (3)

3.2 Land reform

3.2.1 Differences between land redistribution and land restitution

Land redistribution

- Giving previously disadvantaged ✓people access to land ✓

Land restitution

- Returning land ✓ to people who were forcibly removed ✓ (4)

3.2.2 Function of land reform programmes

- To restore land rights ✓ (1)

3.2.3 Models of land ownership in South Africa

- State/government landownership ✓
- Communal landownership ✓
- Freehold land/private land ownership ✓ (3)

3.3 Agricultural organisations

3.3.1 Define the underlined concept

- Agricultural organisation is an institution ✓ consisting of people who are involved in agriculture ✓ (2)

3.3.2 Functions of the Agricultural Research Council (ARC)

- Facilitates enhanced and ensures natural resource conservation ✓
- Sustaining a comparative agricultural economy ✓
- Provides new economic opportunities ✓
- Ensures high-quality and safe food ✓
- Contributes to better quality of life ✓
- Encourages national growth and development of South Africa ✓ (Any 3 x 1) (3)

3.4 Indigenous plants

3.4.1 Identification of the plants in DIAGRAM A and DIAGRAM B

- DIAGRAM A – Aloe ✓
- DIAGRAM B – Hoodia gordonia ✓ (2)

3.4.2 Indication of the plants

(a) Diagram A plant ✓ (1)
(b) Diagram B plant ✓ (1)
(c) Diagram A plant ✓ (1)

3.4.3 Advantages of indigenous farming practices

- Chemical free ✓
- Environmentally friendly ✓
- Low cost ✓
- Non-intensive ✓
- Adaptation to suit local environment ✓ (3)

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3.5 Laws affecting agriculture

3.5.1	(a)	Resource protection laws ✓	(1)
	(b)	Disease and chemical laws ✓	(1)
	(c)	Labour laws ✓	(1)
	(d)	Resource protection laws ✓	(1)
	(e)	Land and land reform laws ✓	(1)
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QUESTION 4: ANIMAL STUDIES

4.1 Nguni cattle breed

4.1.1 Advantages of the breed's thick pigmented hide

- Ticks cannot attach to hide ✓
- Heat tolerant ✓ (2)

4.1.2 Explanation of the term *marbling*

- Meat is streaked with fat ✓ within the lean sections of the meat. ✓ (2)

4.1.3 Importance of the breed in the cultural lives of the African people

- Skins have cultural value ✓
- Lobola ✓ (2)

4.1.4 Identification of a characteristic that enables the Nguni to forage on steep slopes and in thick bushes

- Narrow hooves ✓ (1)

4.2 Intensive farming with broiler chickens

4.2.1 Breeds that would be ideal for intensive broiler production

- Ross ✓
- Cobb ✓ (2)

4.2.2 Basic requirements for successful poultry production

- Housing ✓
- Management ✓
- Breeding ✓
- Nutrition ✓ (4)

4.3 Goat animals

4.3.1 Differences between grazers and browsers

Grazers: Animals that only graze on grass ✓ and other plants close to the ground. ✓

Browsers: Animals that eat plants and shrubs ✓ above ground level ✓ (4)

4.3.2 Goats by-products and their uses

- Hides ✓– used for manufacture of leather goods ✓
- Hair ✓– used in the textile industry to make cashmere ✓
- Horns ✓– used to make spoons ✓
- Intestines ✓ – used to make 'catgut' which is used as material for internal human surgical sutures and strings for musical instruments ✓
- Faeces ✓ – collected for use as an organic fertiliser. ✓

(Any 2 x 2) (4)

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4.3.3 **Define the underlined term**

- Ruminants – animals that have four-part stomach (complex stomach) ✓ that later regurgitate previously swallowed food to chew and swallow again. ✓ (2)

4.4 **Game farming**

4.4.1 **Examples of the big five game animals**

- Rhino ✓
- Lion ✓
- Buffalo ✓
- Leopard ✓
- Elephant ✓ (Any 2 x 1) (2)

4.4.2 **Importance of game farming**

- Contributes to the conservation of biodiversity and wildlife ✓
- Creates job opportunities ✓
- Helps economic growth ✓
- Tourist attraction ✓ (Any 3 x 1) (3)

4.4.3 **Uses of game animals**

- Tourism ✓
- Meat ✓
- Hides ✓
- Trophy hunting ✓ (Any 1 x 1) (1)

4.5 **Illustration of THREE animals**

4.5.1 **Indicate with a suitable reason which of the above animals you would recommend for each of the young farmers mentioned below.**

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(a) Animal A ✓ – Produces large volumes of milk ✓with low butterfat content ✓ (3)

(b) Animal B ✓ – Supplies lactose-free milk ✓ that is ideal for lactose-intolerant people ✓ (3)

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TOTAL SECTION B: 105
GRAND TOTAL: 150

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